Chapter 7 water cycle and watersheds: exam review

Ms. Reyes's Ch. 7 notes; exam Fri Nov 6, 40 questions

If you only read one thing: the water cycle has no starting point, only about 2.5% of Earth's water is fresh, and a watershed is the land that drains to one outlet.

Legend: ★ likely on the exam · ⚠ common mistake · → leads to

FRESHWATER NUMBERS ★

Share	Of what
97%	salt water (oceans)
2.5%	fresh water
69%	of fresh: glaciers, ice caps
30%	of fresh: groundwater
under 1%	of fresh: lakes, rivers, swamps

Only a tiny share of fresh water is easy for people to use.

CYCLE TERMS ★

Term	Meaning
Evaporation	liquid to vapor, solar heat, any temperature
Transpiration	plants release vapor via leaves
Condensation	vapor cools to droplets: clouds
Precipitation	rain, snow, sleet, hail
Runoff	flows over land to streams
Infiltration	soaks into soil; can recharge groundwater

ORDER OF THE CYCLE ★

1. Evaporation
2. Transpiration
3. Condensation
4. Precipitation
5. Runoff
6. Infiltration

Many paths; not one fixed line.

RESIDENCE TIME

Place	About
Atmosphere	9 days
Rivers	2 weeks
Ocean	3,200 years
Glaciers	thousands of years

GROUNDWATER WORDS

Term	Meaning
Aquifer	rock or sediment holding and moving groundwater
Water table	top of fully saturated zone; rises and falls
Recharge	infiltration adds water to aquifer

WATERSHEDS ★

- Land that drains to one outlet (river, lake, sea).
- Boundary: a ridge or high ground, the divide.
- Fertilizer, pollution, paving all reach the outlet.
- Pavement → less infiltration → more runoff → faster stream rise after storms.

`Runoff = Precip − Infiltration − Evaporation`

Ignores plants and storage.

Example (practice numbers): 100 - 30 - 50 = 20 runoff.

MS. REYES SAYS KNOW ★

- The terms.
- The order of the cycle.
- What a watershed is.
- The freshwater numbers.

COMMON MISTAKES ⚠

- ⚠ Transpiration is plants; evaporation is surfaces, because transpiration comes from living leaves.
- ⚠ The water table is not flat, because it follows the land a little.
- ⚠ The cycle does not start at the ocean, because it has no starting point.
- ⚠ Fresh water is about 2.5%, not "most" water, because oceans hold about 97%.